

Semiconductors

Q. What is semiconductor? What is its importance & properties?

① They are not good conductors & not good insulators

② Revolution due to semi conductors in technology — Computers
— Communication & space program — mobile phones
— Laptops

③ In older days
Bulb → LED

④ properties of semiconductor

(A) Can control the charge.

(B) Can provide unidirectional current.

⑤ vacuum Tube

⑥ Types of semiconductor

(A) Intrinsic semiconductor

eg → Si, Ge → Pure

(B) Extrinsic semiconductor

eg → P type } impurity (doping)
N type

→ Disadv ÷ (A) Large size.

(B) Heating is required.

(C) vacuum is required.

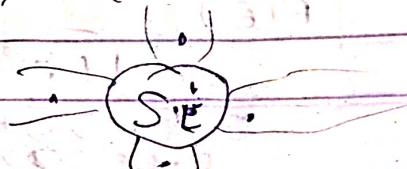
(D) Run at high voltage & power

(E) Less life.

(F) Reliability is less.

Intrinsic semiconductor.

→ Inborn, natural, pure (Si, Ge)



- ① At absolute zero temp, semiconductors are insulators.
- ② Temp. effect (10^{23} atoms - 1 bond is broken)
- ③ Recombination.
- ④ $n_e = n_h = n_i$

⑤ $n_0 = N_c e^{\frac{E_F - E_c}{kT}}$, $n_h = N_v e^{\frac{E_v - E_F}{kT}}$

⑥ Current in semiconductor $I = I_e + I_h$

- ⑦ Drawback (At room temp its conductivity is less because it doesnot break bond freely)

* Derive Expression for density of electrons in conduction band (No. of e^- / conc. of e^-)

$n = N_c e^{\frac{E_F - E_c}{kT}}$

$\rightarrow \int_{E_c}^{E_F} dE \cdot dn$

$dn = \text{Density of states} \times \text{Fermi function}$
 $= Z(E) dE \times F(E)$

$dn = \frac{\pi}{2} \left(\frac{8me^*}{h^2} \right)^{3/2} (E - E_c)^{1/2} \cdot e^{\frac{E_F - E}{kT}} \cdot dE$

$F(E) = \frac{1}{1 + e^{\frac{E - E_F}{kT}}}$

$E = E_F + kT$
 $E = E_F - kT$

$n = \frac{\pi}{2} \left(\frac{8me^*}{h^2} \right)^{3/2} \frac{e^{\frac{E_F}{kT}}}{\int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-\frac{E - E_c}{kT}} dE}$

Let $E - E_c = x$, $E = E_c + x$

$dE = dx$

$dE = 0 = dx$

$\frac{dE}{E = 0, x = 0} = \frac{dx}{x = \infty}$

$n = \frac{\pi}{2} \left(\frac{8me^*}{h^2} \right)^{3/2} \frac{e^{\frac{E_F}{kT}}}{\int_0^{\infty} x^{1/2} e^{-\frac{x}{kT}} dx}$

$n = \frac{\pi}{2} \left(\frac{8me^*}{h^2} \right)^{3/2} e^{\frac{E_F}{kT}} \cdot e^{-\frac{E_c}{kT}} \left[\int_0^{\infty} x^{1/2} e^{-\frac{x}{kT}} dx \right]$
 gamma func.

$n = \frac{\pi}{2} \left(\frac{8me^*}{h^2} \right)^{3/2} e^{\frac{E_F - E_c}{kT}} \times (kT)^{3/2} \frac{\sqrt{\pi}}{2}$

$n = \frac{1}{4} \left(\frac{8\pi me^* kT}{h^2} \right)^{3/2} \cdot e^{\frac{E_F - E_c}{kT}}$

$n = \left[\frac{2(2\pi me^* kT)}{h^2} \right]^{3/2} e^{\frac{E_F - E_c}{kT}}$

Pauli exclusion

N_c

$n_e = N_c e^{\frac{E_F - E_c}{kT}}$

Q Derive exp for density of hole in valence band (No. of holes / conc.)

$$n_h = N_v e^{\frac{E_v - E_F}{kT}}$$



$$Z(E) dE = \frac{\pi}{2} \left(\frac{8 m_h^*}{h^2} \right)^{3/2} \frac{1}{E} dE$$

$$Z(E) = \frac{\pi}{2} \left(\frac{8 m_h^*}{h^2} \right)^{3/2} (E_v - E)^{1/2}$$

$$\Rightarrow 1 - f(E) = 1 - \frac{1}{1 + e^{\frac{E - E_F}{kT}}}$$

$$= \frac{1 - 1}{1 + e^x} = \frac{-1 + 1}{1 + e^x} = \frac{e^x}{1 + e^x}$$

$$= \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-\frac{E - E_F}{kT}}}$$

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$$n_H = \frac{\pi}{2} \left(\frac{8 m_h^*}{h^2} \right)^{3/2} e^{-E_F/kT} e^{E/kT} \int_0^\infty n^{1/2} e^{-x/kT} (-dn)$$

$$n_H = \frac{\pi}{2} \left(\frac{8 m_h^*}{h^2} \right)^{3/2} e^{\frac{E_v - E_F}{kT}} \int_0^\infty n^{1/2} e^{-x/kT} dx$$

$$n_H = \frac{\pi}{2} \left(\frac{8 m_h^*}{h^2} \right)^{3/2} e^{\frac{E_v - E_F}{kT}} (kT)^{3/2} \int_0^\infty \frac{1}{2} e^{-x} dx$$

$$n_H = \frac{1}{4} \left(\frac{8 \pi m_h^* kT}{h^2} \right)^{3/2} e^{\frac{E_v - E_F}{kT}}$$

$$n_H = \left(\frac{2 \pi m_h^* kT}{h^2} \right)^{3/2} e^{\frac{E_v - E_F}{kT}}$$

$$n_H = N_v e^{\frac{E_v - E_F}{kT}}$$

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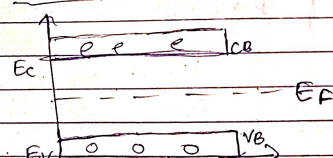
$$n_H = N_v e^{\frac{E_v - E_F}{kT}}$$

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$$n_H = N_v e^{\frac{E_v - E_F}{kT}}$$

Fermi level in Intrinsic semiconductor



$$\Rightarrow \textcircled{1} n_e = n_h$$

$$N_c e^{\frac{(E_F - E_c)}{kT}} = N_v e^{\frac{(E_v - E_F)}{kT}}$$

$$2 \left(\frac{2 \pi m_e^* kT}{h^2} \right)^{3/2} e^{\frac{(E_F - E_c)}{kT}} = 2 \left(\frac{2 \pi m_h^* kT}{h^2} \right)^{3/2} e^{\frac{(E_v - E_F)}{kT}}$$

$$(m_e^*)^{3/2} e^{(E_F - E_C)/KT} = (m_h^*)^{3/2} e^{(E_V - E_F)/KT}$$

$$e^{(2E_F/KT)} = \left(\frac{m_h^*}{m_e^*}\right)^{3/2} e^{(E_V + E_C)/KT}$$

$$\frac{2E_F}{KT} \ln e = \ln \left(\left(\frac{m_h^*}{m_e^*}\right)^{3/2} e^{(E_V + E_C)/KT} \right)$$

$$\frac{2E_F}{KT} = \frac{3}{2} \ln \left(\frac{m_h^*}{m_e^*} \right) + \frac{(E_V + E_C)}{KT} \ln e$$

$$\frac{2E_F}{KT} = \frac{3}{2} \ln 1 + \frac{E_V + E_C}{KT}$$

$$\frac{2E_F}{KT} = \frac{E_V + E_C}{KT}$$

$$\frac{2E_F}{KT} = \frac{E_V}{KT} + \frac{E_C}{KT}$$

$$E_F = \left(\frac{E_V \times KT}{2} \right) + \left(\frac{E_C \times KT}{2} \right)$$

$$E_F = \frac{E_V + E_C}{2}$$

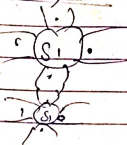
Intrinsic semiconductor

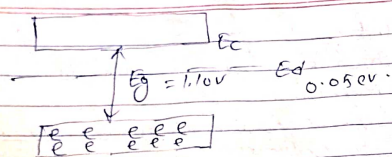
- ① The semiconductor in which impurities are added is called extrinsic semiconductor.
- ② When impurities are added to intrinsic semiconductor, it becomes extrinsic semiconductor.
- ③ The process of adding impurity is called doping.
- ④ Due to doping, created semiconductor is called doped semiconductor.
- ⑤ The impurity is called dopant.
- ⑥ Doping increases the electrical conductivity of semiconductor.
- ⑦ Type of impurity decides the nature of semiconductor.
- ⑧ Extrinsic semiconductor has higher electrical conductivity than intrinsic semiconductor.
- ⑨ They are used in manufacturing of electronic devices (diode).
- ⑩ The number of free electrons and holes are not same in extrinsic semiconductor.

N-Type

- 5 valence e⁻ → Pentavalent impurity → P, Sb, As
- p-type
- 3 valence e⁻ → Trivalent → B, In, Ga, Al

N-Type





- ① Negative type semiconductor
- ② Pentavalent impurity (P, Sb, As)

Si ($Z=14$) = 2, 8, 4 } Conductivity
 P ($Z=15$) = 2, 8, 5 } 16 times ↑

10^8 atom Si - 1 atom P

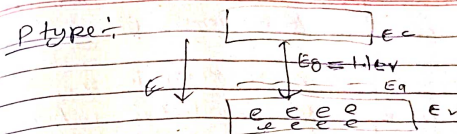
③ Not depending on temperature.

④ $n_e > n_h$

⑤ donor e^- 50%, conduction e^- 10%.

→ majority charge carriers are e^- and minority charge carriers are holes.

→ with the help of pentavalent impurity N type semiconductor is constructed.



① positive type semiconductor

② Trivalent impurities (B, Al, In, Ga)

Si ($Z=14$) = 2, 8, 4 } conductivity
 Al ($Z=13$) = 2, 8, 3 } 16 times ↑

10^8 atom Si - 1 atom Al

③ Not depending on temperature.

④ Electrical conductivity is more.

⑤ $n_h > n_e$.

⑥ Acceptor holes 90%, valence hole 10%.

→ majority charge carriers are holes and minority charge carriers are electrons.

→ with the help of trivalent impurity p type semiconductor is constructed.

Intrinsic semiconductor

- Pure semiconductor
- Electrical conductivity is low
- Density of electron is equal to density of hole
- Depends on temperature only
- No impurities
- Fermi level lies in b/w valence and conduction band

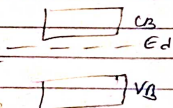
Extrinsic semiconductor

Impure semiconductor

- Electrical conductivity is high
- Density of electron is not equal to density of hole
- Depends on amount of doping
- Trivalent (Al, B, In, Ga) and pentavalent (P, Sb, As) impurities
- Fermi level lies near valence band in P-type & near conduction band in N-type

N-Type

- ① Pentavalent impurities are added e.g. P, Sb, As
- ② One electron comes extra
- ③ Donor impurity
- ④ Donor energy level is present

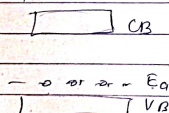


- ⑤ Fermi level is present near to the conduction band

- ⑥ majority charge carriers are e^-
- ⑦ minority charge carriers are holes

P-type

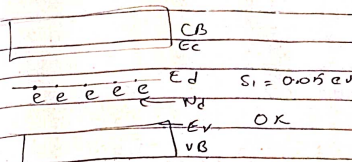
- ① Trivalent impurities are added e.g. Al, In, B, Ga
- ② One hole comes extra
- ③ Acceptor impurity
- ④ Acceptor energy level is present



- ⑤ Fermi level near to valence band

- ⑥ majority charge carriers are holes
- ⑦ minority charge carriers are e^-

Fermi level in N-type semiconductor (dependence of Fermi level on temp)



$$n_d = 2 \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/2} e^{\frac{(E_F - E_c)}{kT}} \quad \text{--- (1)}$$

$$N_d = N_d^+ \quad \text{--- (2)}$$

$$N_d^+ = \text{Total} \times \text{probability (+ve charge)} \\ = N_d \times [1 - f(E)]$$

$$= N_d \left[1 - \frac{1}{1 + e^{\frac{(E - E_F)}{kT}}} \right]$$

$$\frac{E_d - E_F}{kT} = \ln x$$

$$= N_d \left[1 - \frac{1}{1 + e^x} \right]$$

$$N_d^+ = N_d \left[\frac{1 + e^x - 1}{1 + e^x} \right]$$

$$= N_d \left[\frac{e^x}{1 + e^x} \right] = N_d \left[\frac{1}{1 + e^{-x}} \right] = N_d \left[\frac{1}{1 + \frac{e^{-x}}{e^x}} \right]$$

$$= N_d \left[\frac{1}{1+e^n} \right] = N_d \left[\frac{1}{1+e^{-n}} \right]$$

$$= N_d \left[\frac{1}{1+e^{\frac{(E_F - E_d)}{KT}}} \right]$$

$$= N_d \left[\frac{1}{e^{\frac{(E_F - E_d)}{KT}}} \right] \quad \left\{ \begin{array}{l} E_F - E_d \gg KT \\ \frac{E_F - E_d}{KT} \gg 1 \end{array} \right. \therefore e^{\frac{(E_F - E_d)}{KT}} \gg 1$$

$$N_d^+ = N_d \cdot e^{\frac{E_d - E_F}{KT}} \quad (3) \quad \text{neglecting 1}$$

$$eq^n \text{ (1) } - (3)$$

$$2 \left(\frac{2\pi m_e^* KT}{h^2} \right)^{3/2} e^{\frac{(E_F - E_c)}{KT}} = N_d \cdot e^{\frac{(E_d - E_F)}{KT}}$$

$$\therefore e^{\frac{2E_F - E_c - E_d}{KT}} = \frac{N_d}{2} \left(\frac{h^2}{2\pi m_e^* KT} \right)^{3/2}$$

$$\therefore e^{\frac{2E_F - (E_c + E_d)}{KT}} = \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}$$

→ Taking ln on both sides

$$\therefore \frac{2E_F - (E_c + E_d)}{KT} = \ln \left[\frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2} \right]$$

$$\therefore \frac{2E_F}{KT} - \frac{E_c + E_d}{KT} = \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}$$

$$\therefore \frac{2E_F}{KT} - \frac{E_c + E_d}{KT} = \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}$$

$$\therefore \frac{2E_F}{KT} = \frac{E_c + E_d}{KT} + \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}$$

$$\therefore E_F = \frac{E_c + E_d}{2} + \frac{KT}{2} \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}$$

At $T=0K$.

$$\left[E_F = \frac{E_c + E_d}{2} \right]$$

On increasing the temperature, Fermi level will be shifted towards the valence band.

• Charge concentration in conduction band for N-type semiconductor

$$n_e = 2 \left(\frac{2\pi m_e^* KT}{h^2} \right)^{3/2} e^{\frac{E_F - E_c}{KT}} \quad (1)$$

$$E_F = \frac{E_c + E_d}{2} + \frac{KT}{2} \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2} \quad (2)$$

$$e^{\frac{E_F - E_c}{KT}} = e^{\frac{E_F}{KT}} \cdot e^{-\frac{E_c}{KT}}$$

$$= e^{\frac{E_c + E_d}{2} + \frac{KT}{2} \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}} \cdot e^{-\frac{E_c}{KT}}$$

$$= e^{\frac{E_c + E_d}{2KT}} \cdot e^{\frac{1}{2} \ln \frac{N_d}{2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}} \cdot e^{-\frac{E_c}{KT}}$$

$$= e^{\frac{E_c + E_d - E_c}{2KT}} \cdot e^{\ln \left(\frac{N_d}{2} \right)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2} \right)^{-3/2}}$$

$$= e^{\frac{E_C + E_D - 2E_C}{2KT}} \cdot \left(\frac{N_D}{2}\right)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2}\right)^{-3/4}$$

$$e^{\frac{E_F - E_C}{KT}} = e^{\frac{E_D - E_C}{2KT}} \left(\frac{N_D}{2}\right)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2}\right)^{-3/4}$$

$$n_e = 2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{3/4} e^{\frac{E_D - E_C}{2KT}} \left(\frac{N_D}{2}\right)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2}\right)^{-3/4}$$

$$\therefore n_e = 2 \left(\frac{2\pi m_e^* KT}{h^2}\right)^{3/4} \left(\frac{N_D}{2}\right)^{1/2} e^{\frac{E_D - E_C}{2KT}}$$

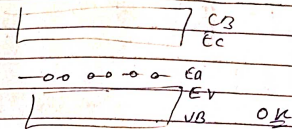
$$n_e = 2^{1/2} \left(\frac{N_D}{2}\right)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2}\right)^{3/4} e^{\frac{E_D - E_C}{2KT}}$$

$$n_e = 2^{1/2} \times 2^{-1/2} \times (N_D)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2}\right)^{3/4} e^{\frac{E_D - E_C}{2KT}}$$

$$n_e = (2 N_D)^{1/2} \left(\frac{2\pi m_e^* KT}{h^2}\right)^{3/4} e^{\frac{E_D - E_C}{2KT}}$$

Charge density in cs } N type semiconductor
Number of e^- in cs

Fermi level in P type semiconductor :-
(dependence of fermi level on temp)



$$n_h = 2 \left(\frac{2\pi m_h^* KT}{h^2}\right)^{3/4} e^{\frac{(E_V - E_F)}{KT}} \quad (1)$$

$$N_A = N_A^- \quad (2)$$

N_A^- = Total $\times$ probability (-ve charge)

$$= N_A \times f(E)$$

$$= N_A \left(\frac{1}{1 + e^{\frac{(E - E_F)}{KT}}} \right)$$

$$N_A^- = N_A \left(\frac{1}{1 + e^{\frac{(E_A - E_F)}{KT}}} \right)$$

$$N_A^- = N_A e^{\frac{(E_F - E_A)}{KT}} \quad (3)$$

$$E_F = E_V + E_A$$

$$E_A - E_F > KT$$

$$\frac{E_A - E_F}{KT} \gg 1$$

$$\therefore e^{\frac{(E_A - E_F)}{KT}} \gg 1$$

$\rightarrow$ neglecting (1)

$$(1) = (2)$$

$$\therefore 2 \left(\frac{2\pi m_h^* KT}{h^2}\right)^{3/4} e^{\frac{(E_V - E_F)}{KT}} = N_A e^{\frac{(E_F - E_A)}{KT}}$$

$$e^{\frac{(E_V - 2E_F + E_A)}{KT}} = \frac{N_A}{2} \left(\frac{h^2}{2\pi m_h^* KT} \right)^{3/4}$$

$$e^{\frac{E_V - 2E_F + E_G}{KT}} = \frac{N_A}{2} \left(\frac{h^2}{2\pi m_A^* kT} \right)^{3/2}$$

$$e^{E_V - 2E_F + E_G} = \frac{N_A}{2} \left(\frac{2\pi m_A^* kT}{h^2} \right)^{-3/2}$$

Taking ln both sides

$$\frac{E_V - 2E_F + E_G}{KT} \ln e = \ln \left[\frac{N_A}{2} \left(\frac{2\pi m_A^* kT}{h^2} \right)^{-3/2} \right]$$

Multiply (-1) on both side

$$\frac{2E_F - E_V - E_G}{KT} = -\ln \left[\frac{N_A}{2} \left(\frac{2\pi m_A^* kT}{h^2} \right)^{-3/2} \right]$$

$$\frac{2E_F - (E_V + E_G)}{KT} = -\ln \left(\frac{N_A}{2} \right) \left(\frac{2\pi m_A^* kT}{h^2} \right)^{-3/2}$$

$$\frac{2E_F}{KT} - \frac{E_V + E_G}{KT} = -\ln \left(\frac{N_A}{2} \right) \left(\frac{2\pi m_A^* kT}{h^2} \right)^{-3/2}$$

$$\therefore E_F = \frac{E_V + E_G}{2} - \frac{KT}{2} \ln \left(\frac{N_A}{2} \right) \left(\frac{2\pi m_A^* kT}{h^2} \right)^{-3/2}$$

at $T=0K$

$$E_F = \frac{E_V + E_G}{2}$$

on increasing the temperature fermi level is shifted towards conduction band.

Change concⁿ in valence band for p-type semiconductor.

$$n_h = \frac{2(2\pi m_e^* kT)}{h^2} e^{\frac{E_V - E_F}{KT}} \quad \text{--- (1)}$$

$$E_F = \frac{E_V + E_G}{2} - \frac{KT}{2} \ln \frac{N_A}{2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/2} \quad \text{--- (2)}$$

$$e^{\frac{E_V - E_F}{KT}} = e^{\frac{E_V}{KT}} \cdot e^{-\frac{E_F}{KT}}$$

$$= e^{\frac{E_V}{KT}} \cdot e^{-\left[\frac{E_V + E_G}{2} - \frac{KT}{2} \ln \frac{N_A}{2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/2} \right] \frac{1}{KT}}$$

$$= e^{\frac{E_V}{KT}} \cdot e^{-\frac{E_V + E_G}{2KT}} + \frac{1}{2} \ln \frac{N_A}{2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/2}$$

$$= e^{\frac{E_V}{KT} - \frac{E_V + E_G}{2KT}} \cdot e^{\frac{1}{2} \ln \frac{N_A}{2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/2}}$$

$$= e^{\frac{2E_V - E_V - E_G}{2KT}} \cdot e^{\frac{1}{2} \ln \frac{N_A}{2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/2}} \quad \text{--- (3)}$$

$$e^{\frac{E_V - E_F}{KT}} = e^{\frac{E_V - E_G}{2KT}} \cdot \left(\frac{N_A}{2} \right)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/4}$$

$$n_h = \frac{2(2\pi m_e^* kT)}{h^2} \cdot e^{\frac{E_V - E_G}{2KT}} \left(\frac{N_A}{2} \right)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{-3/4}$$

$$n_h = \frac{1}{2} \cdot 2 \left(\frac{N_A}{2} \right)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/2 - 3/4} e^{\frac{E_V - E_G}{2KT}}$$

$$n_h = 2^{1/2} \times 2^{-1/2} (N_A)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/4} e^{\frac{E_V - E_G}{2KT}}$$

$$n_n = 2^{1/2} (N_A)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/4} e^{\frac{E_V - E_g}{2kT}}$$

$$n_h = (2N_A)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/4} e^{\frac{E_V - E_g}{2kT}}$$

Charge density in VB } p-type
no. of holes in VB

Law of Mass action

$$n \cdot p = n_i^2 = \text{const. (fixed temp.)}$$

No. of e⁻ in conduction band No. of holes in VB

$$n_n \cdot p_n = n_i^2 \rightarrow \text{N-type}$$

$$p_p \cdot n_p = n_i^2 \rightarrow \text{P-type}$$

Generation of Carriers

- The process by which thermally free e⁻s and holes are generated in pairs is called generation of carriers.
- When electrons in valence band get enough energy then they will absorb this energy & jump into the conduction band.
- The electron which is jumped into conduction

band is called thermally generated free electron and the place from where electron left is called hole. So, two types of carriers are generated (1) electron (2) hole.

(2) Recombination of carriers

→ The process of filling back of thermally generated free e⁻ from conduction band to the hole present in valence band is called as recombination of carriers.

→ In this process, thermally generated free electron from conduction band to the hole present in the valence band is called as recombination of carriers.

→ In this process, thermally generated free e⁻ and holes are eliminated.

⇒ Law of mass action:

'product of number of electrons in conduction band & number of holes in valence band is constant at fixed temperature and is independent of amount of donor and acceptor impurities added.'

$$n \cdot p = n_i^2 = \text{const.}$$

* Extrinsic semiconductor:

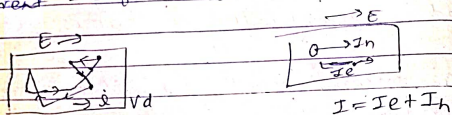
• product of majority charge carriers & minority charge carriers is constant at fixed temperature and is independent of amount of donor and acceptor impurity added.

(A) N-type : $n_n p_n = n_i^2 = \text{const}$

(B) P-type : $p_p n_p = n_i^2 = \text{const}$

Drift and diffusion current

Total current = drift + diffusion



$$I = v_d e n A$$

$$v_d = \mu E$$

$$I_e = \mu_e E e n A$$

$$I_h = \mu_h E e n A$$

$$\frac{I_e}{A} = \mu_e E e n$$

$$\frac{I_h}{A} = \mu_h E e n$$

$$J_e = \mu_e E e n$$

$$J_h = \mu_h E e n$$

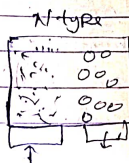
$$J = \mu_e E e n + \mu_h E e n$$

$$\therefore J = J_e + J_h$$

$$J = e n (\mu_e + \mu_h) E$$

$$J_e = e \left(D_e \frac{dn}{dx} \right) \quad \text{diffusion current constant}$$

$$J_h = -e D_h \frac{dp}{dx}$$



$$J = J_e + J_h$$

$$J_e = \mu_e E e n + e D_e \frac{dn}{dx}$$

$$J_h = \mu_h E e n - e D_h \frac{dp}{dx}$$

$$J = J_e + J_h$$

mobility μ

Electric force on an electron $F = qE$

Newton's second law, $F = ma$

$$a = \frac{F}{m}$$

$$a = \frac{qE}{m} \text{ for one electron.}$$

putting for all electrons

$$\int a dt = \int \frac{qE}{m} dt$$

$$\int \frac{dv}{dt} dt = \int \frac{qE}{m} dt$$

$$\int dv = \frac{qE}{m} \int dt + C$$

$$v = \frac{qEt}{m} + C \quad \text{--- (1)}$$

when $t=0$, $v=0$

$$0 = \frac{qE}{m} (0) + C$$

$$C = 0$$

putting value of c in eqⁿ (1)

$$v = \frac{qE}{m} t$$

$$v = \left(\frac{qE}{m} \right) t$$

$$\left[\frac{v \propto E}{\text{avg. val.}} \right] \cdot \frac{E}{E} \\ \boxed{v = \mu E}$$

$$\boxed{\mu = \frac{v}{E}}$$

$$S.I : \frac{m^2}{\text{volt} \cdot \text{sec}}$$

"mobility of charge carrier is defined as its average velocity under the influence of a unit electric field strength".

$$\text{① } \overrightarrow{e^-}$$

$$\rightarrow E = 50 \text{ N/C}$$

Effective mass of an electron (m_e)

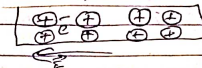
① e^- is present in vacuum

$$e^- \rightarrow F_E$$

$$F_E = qE$$

$$F_E = eE$$

② e^- is present inside the crystal



$$F = F_E + F_n$$

$$m = e^- \text{ (when } e^- \text{ is present in vacuum)}$$

$$F_n = m_1 a$$

$$m_1 = e^- \text{ (where } e^- \text{ is present in crystal)}$$

$$F = F_E + F_n$$

$$m a = eE + m_1 a$$

$$m a - m_1 a = eE$$

$$m - m_1 = \frac{eE}{a}$$

$$\boxed{m^* = \frac{F_{ext}}{a}}$$

P-N Juncⁿ & PN Juncⁿ Diode.

→ PN Junction: The boundary separated P-type & N-type semiconductor is known as PN Juncⁿ.

→ PN Junction Diode: By connecting P-type & N-type semiconductor, prepared device is PN Junction diode.

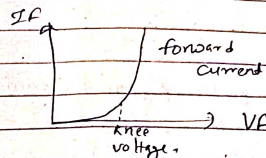
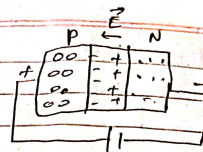
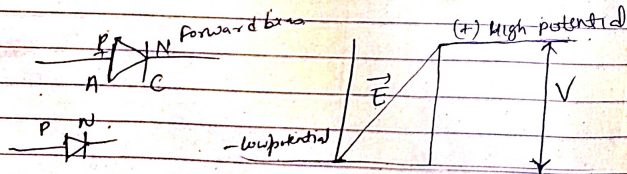
→ Diffusion current → momentary motion of electron creates diffusion current.

→ Diffusion process → The process in which diffusion current is produced.

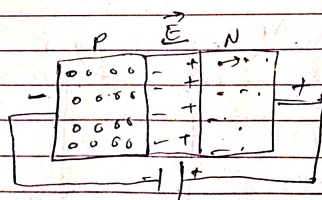
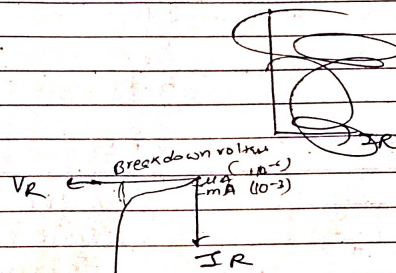
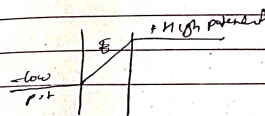
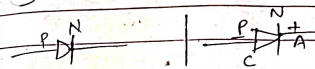
→ Depletion layer → Such layer where shortage of charge is present due to diffusion process.

→ potential barrier: The E field set up in depletion layer creates potential V according to $V = E \cdot d$. This potential opposes the flow of current. So it is known as potential barrier. $S_i = 0.6 \text{ eV}$, $W_e = 0.3 \text{ eV}$

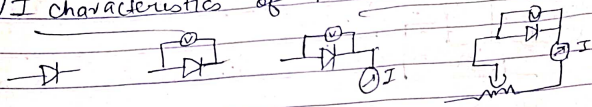
Forward bias P-N Juncⁿ Diode



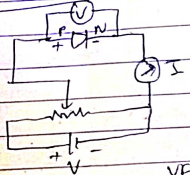
(1) Reverse bias P-N Juncⁿ Diode



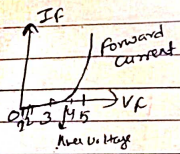
V-I characteristics of PN Junction Diode



V-I characteristics of forward bias

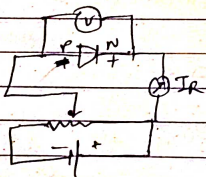


max ~~current~~ forward current



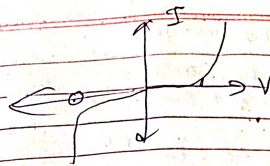
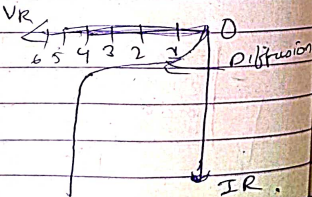
V_F	I_F
0	0
1	0
2	0
3	0
4	1
5	2

V-I char. of reverse bias



V_R	I_R
0	0
1	1 μ A
2	1 μ A
3	1 μ A
4	10mA
5	20mA
6	30mA

Peak Inverse Voltage (PIV)

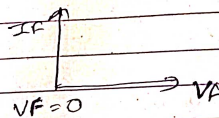


Ideal diode (1st approximation of PN Junction Diode)

Forward bias (practical diode)

Potential barrier = 0

- Completely Allows the current
- provide zero resistance
- Acts as a perfect conductor

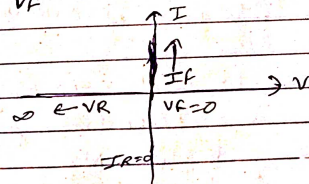
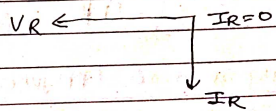


Reverse bias

open ext. openswitch

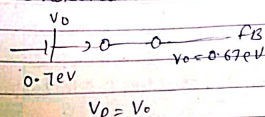
Potential barrier = ∞

- Completely blocks the current
- provide infinite resistance
- Acts as a perfect insulator



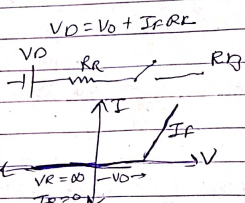
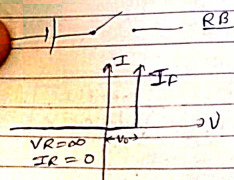
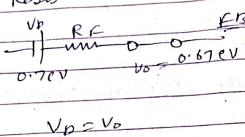
2nd approximation

potential barrier is considered (0.67eV)



3rd approximation

potential barrier and Resistance are considered



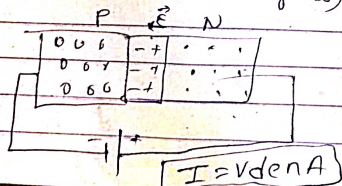
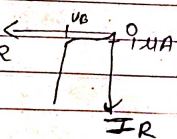
Zener Diode $\div$ ($-P-N+$ $\rightarrow$ Reverse bias)

$\rightarrow$ special purpose diode (PN junction diode)

$\rightarrow$ used as voltage regulator

① forward characteristics (+, +), (-, -), correct connection
current flows

② reverse characteristics (-, +), (+, -) wrong connection
no current flows

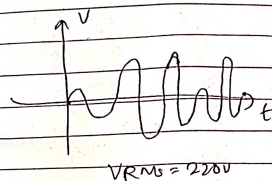


$$I = V_D / R_A$$

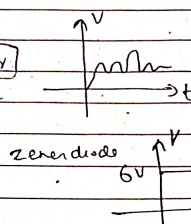
* Zener effect: Impurity $\uparrow \rightarrow$ Depletion layer $\uparrow \rightarrow$ E-field $\uparrow$
 $\rightarrow$ EET $\rightarrow$ Breaking of bond $\rightarrow$ nT (At V_Z)

* Avalanche effect: Impurity $\downarrow \rightarrow$ Depletion layer $\uparrow \rightarrow$ E-field in p-type $\rightarrow$ Breaking bond $\rightarrow$ nT (At V_{AK})

Zener diode as a voltage Regulator



Rectifier
AC $\rightarrow$ DC

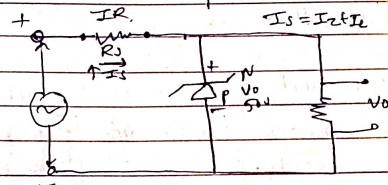


KVL

$$V_i - I_S R_S - V_0 = 0$$

$$V_i = I_S R_S + V_0$$

$$V_i = (I_Z + I_L) R_S + V_0$$



Use

① As a voltage Regulator

② As a reference voltage device

③ constant current source with transistor

④ over voltage protection ⑤ clipping & clamping circuit

Testing of diode with multimeter

① Terminals

- P type (Anode-A)
- N type (Cathode-C)

② Biasing

- forward bias ($I = I_R$)
- Reverse bias ($I = 0$, R_T)

* Condⁿ

- ok (healthy)
- faulty.

$$\frac{P}{A} \frac{D}{L} \frac{N}{C}$$

Schottky diode

PN Junction diode

- Depletion layer is high
- E is high
- potential barrier is high (0.6 eV)
- forward potential drop is high
- switching action is poor (at high freq)
- Bipolar device

Schottky diode

- Depletion layer is negligible
- E is negligible
- potential barrier is low (0.3 eV)
- forward potential drop is less
- switching action is quick (at high freq)

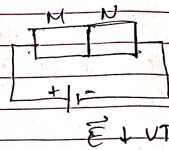
→ unipolar device

→ hot electron diode

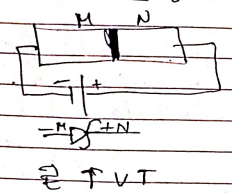
Schottky diode



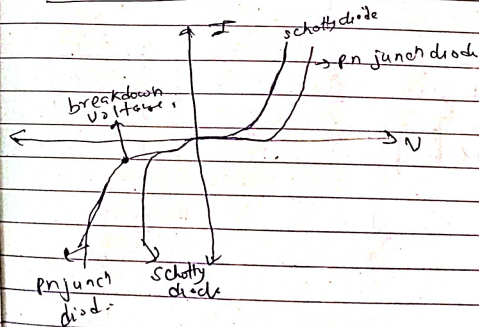
* forward biasing



* Reverse biasing



* V-I Characteristics



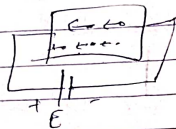
Mobility and Resistivity

mobility (μ) $\div$ It measures the ease with which carriers can move within the crystal.

$$V_d \propto E$$

$$V_d = \mu_n E$$

$$\mu_n = \frac{V_d}{E} = \frac{m^2}{V \cdot s}$$



Resistivity (ρ) $\div$ It is the property of a material that measures how strong the material opposes the flow of current at a particular temperature.

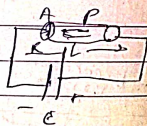
$$J = \frac{I}{A}$$

$$J = q n v_d A \Rightarrow J = q n v_d$$

$$J = q n \mu_n E = \sigma E$$

conductivity

$$\sigma = q n \mu_n$$



$$\sigma = \frac{1}{\rho} \Rightarrow \rho = \frac{1}{\sigma}$$

$$\rho = \frac{1}{q n \mu_n}$$

Poisson's continuity Equation

Statement $\div$ An equation that expresses the equality of incoming and outgoing charges in a system and follows the law of conservation of charges is called Poisson's continuity eqn.

Physical significance $\div$ It implies that in absence of source or sink of charge the rate of increase of charge in a conductor equals to rate of flow of charge into the conductor. continuity eqn represent charge conservation.

Derivation $\div$ $q = \oint \rho dV$

$$I = - \frac{d q}{dt} = - \frac{d}{dt} \oint \rho dV \quad \text{--- (1)}$$

If $\vec{J}$ = current density then

$$I = \oint \vec{J} \cdot d\vec{s} \quad \text{--- (11)}$$

from eqn (1) and (11)

$$\oint \vec{J} \cdot d\vec{s} = - \frac{dq}{dt} = - \frac{d}{dt} \oint \rho dV$$

using divergence theorem

$$\oint \vec{J} \cdot d\vec{s} = \int \nabla \cdot \vec{J} dV \quad \text{--- (2)}$$

using (3)

$$\int \nabla \cdot \vec{J} dV = - \frac{d}{dt} \oint \rho dV$$

So, $\nabla \cdot \vec{J} = -\frac{d\rho}{dt}$

$\nabla \cdot \vec{J} + \frac{d\rho}{dt} = 0$

as charge inside a conductor is zero, so $\rho = 0$

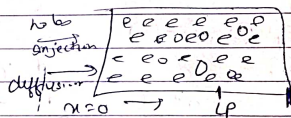
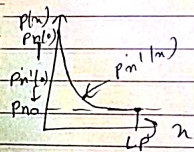
$\nabla \cdot \vec{J} = 0$

Diffusion length and mean lifetime

P_{n0} → Initial conc. of holes in n-type due to thermal excitation.

$P_{n0} = \frac{n_i^2}{N_D}$

L_p → diffusion length



$P_n(0) = P_n'(0) + P_{n0}$

Total at $x=0$

Injection

Thermal excitation

$P_n(x) = P_n'(0)e^{-x/L_p}$

$L_p = \sqrt{D_p \tau_p}$

Avg. lifetime of holes

$P_n(x) = P_{n0} + P_n'(x)$

$P_n(x) = P_{n0} + P_n'(0)e^{-x/L_p}$

$D_p \rightarrow \frac{\text{cm}^2}{\text{s}}$
||
cm

→ Injection (diffusion) → flow of diffusion current.

$J_p(\text{diff}) = -e D_p \frac{dP_n(x)}{dx} \rightarrow P_n(x) = P_{n0} + P_n'(0)e^{-x/L_p}$

$\frac{dP_n(x)}{dx} = \frac{-1}{L_p} P_n'(0)e^{-x/L_p}$

$\frac{dP_n(x)}{dx} = \frac{-1}{L_p} P_n'(x)$

$J_p(\text{diff}) = -e D_p \left(\frac{-1}{L_p} P_n'(x) \right)$

$J_p(\text{diff}) = \frac{e D_p P_n'(x)}{L_p}$ A/cm²

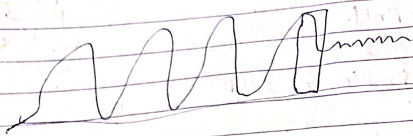
$J_n(\text{diff}) = \frac{-e D_n n' P(x)}{L_n}$

$L_p = \sqrt{D_p \tau_p}$
 $L_n = \sqrt{D_n \tau_n}$

Tunnelling effect/process

- The quantum tunnelling effect allows a confined particle within a finite potential well to penetrate through the classically impenetrable potential wall.

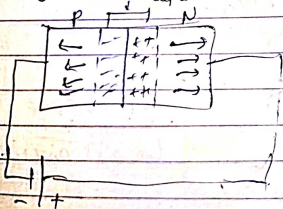
Incident + Reflected wave = Transmitted wave



Capacitance of P-n junction diode
 $C = \frac{A \epsilon}{d}$
 $A \rightarrow$ Area
 $\epsilon \rightarrow$ dielectric constant
 $d \rightarrow$ depletion

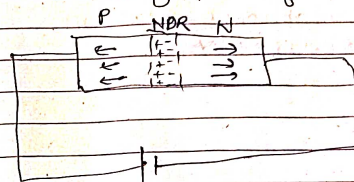
Transition capacitance: During reverse Bias. In R.B.P. and 'N' type region have low resistance. Hence they act like conducting plates of capacitor. The depletion region has high resistance, hence acts like insulating material.

→ parallel plate capacitor charge is stored in depletion region in form of electric field.

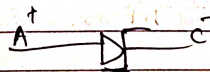


Diffusion capacitance (C_D): During forward Bias, occurs due to stored charge of minority e^- s and holes near the depletion region, hence also called as storage capacitance.

→ depletion region acts like dielectric or insulator and charge stored at both sides of depletion layer acts like conducting plate of the capacitor.



Gunn Diode. → A Gunn diode also known as a transferred electron device (TED) is a form of diode, a two-terminal semiconductor electronic component, with negative differential resistance, used in high frequency electronics. It is based on the Gunn effect. Its main uses are in electronic oscillators to generate microwaves, in applications such as radar speed guns, microwave relay data link transmitters etc.



A light-emitting diode is a semiconductor device that emits light when current flows through it. e^- in the semiconductor recombine with e^- holes, releasing energy in the form of photons.

LED



Light emitting diode (Electrical energy $\rightarrow$ light)

photo diode

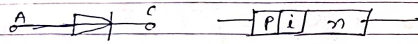


(Light energy $\rightarrow$ Electrical energy (current))

A photodiode is a semiconductor device with a p-n junction that converts photons into electrical current. The p-layer has an abundance of holes and the n-layer has an abundance of electrons.



PIN Diode



A pin diode is a diode with a wide, undoped intrinsic semiconductor region between a p-type semiconductor and an n-type semiconductor region. The p-type and n-type regions are typically heavily doped because they are used for ohmic contacts.

~~Variable Diode~~ The wide intrinsic region is in contrast to an ordinary p-n diode. The wide intrinsic region makes the pin diode an inferior rectifier. (one typical function of a diode), but it makes it suitable for attenuators, fast switches, photo detectors and high voltage power electronic applications, RF and microwave switches

ex $\rightarrow$ SFH 203, BPW 34

LASER DIODE (LD) also Injection Laser diode or ILD or semiconductor laser or diode laser is a semiconductor device similar to a light emitting diode in which a diode pumped directly with electrical current can create lasing conditions at the diode's junction. It is used in Telecommunications, scanning and spectrometry, maskless photolithography



TUNNEL diode

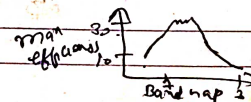


A tunnel diode or Esaki diode is a type of semiconductor diode that has effectively negative resistance due to the quantum mechanical effect

called Tunneling. Tunnel diodes have a heavily doped p-n junction that is about 100nm wide. The heavily doping results in a broken bandgap, where conduction band electron states on the n-side are more or less aligned with valence band holes states on the p-side. They are usually made from germanium, but can also be made from gallium arsenide and silicon materials.

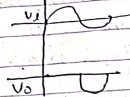
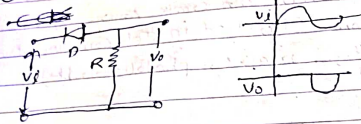
SOLAR CELL or photovoltaic cell is an electronic device that

converts the energy of light directly into electricity by means of the photovoltaic effect. It is a form of photoelectric cell, a device whose electrical characteristics vary when it is exposed to light. Individual solar cell devices are often the electrical building blocks of photovoltaic modules, known colloquially as 'solar panels'. Almost all commercial PV cells consist of crystalline silicon 'ex-cells', modules, panels and systems

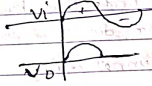
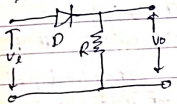


clipping (remove unwanted wave form)

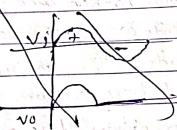
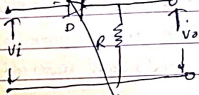
① series positive clipper



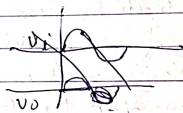
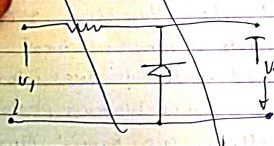
② series Negative clipper



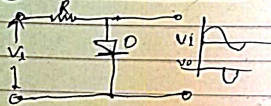
③ shunt positive clipper



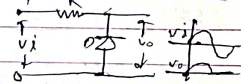
④ shunt negative clipper



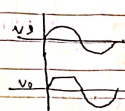
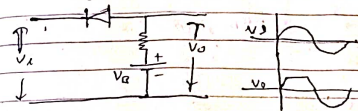
⑤ shunt positive clipper



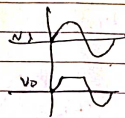
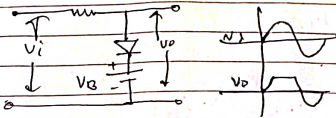
⑥ shunt negative clipper



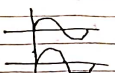
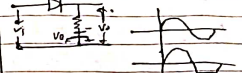
⑦ series biased +ve clipper



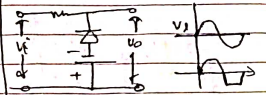
⑧ shunt biased +ve clipper



⑨ series biased -ve clipper



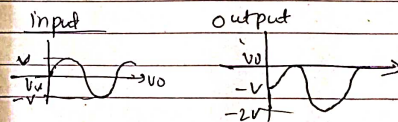
⑩ shunt biased -ve clipper



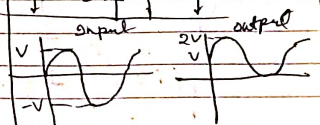
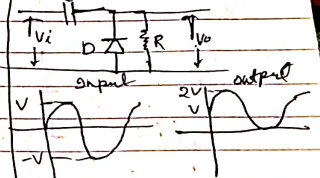
Diode Clamper circuit

→ shift the DC level of signal → Diode, Resistor, Capacitor

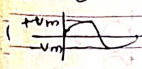
① -ve clamper:



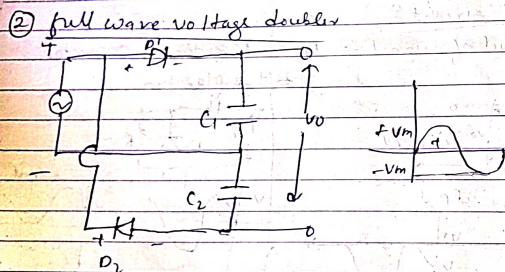
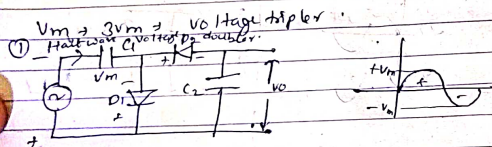
② +ve clamper:



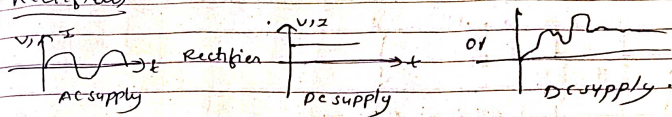
Voltage multipliers

1.  voltage multiplier $\rightarrow V \rightarrow 2V_m, 3V_m, 4V_m$
AC $\rightarrow$ DC form

$V_m \rightarrow 2V_m$ - voltage doubler
(1) half wave voltage doubler
(2) full wave voltage doubler



Rectifiers

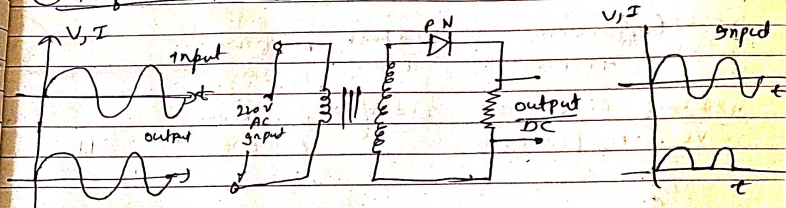


The device which converts AC to DC supply is called 'Rectifier'.

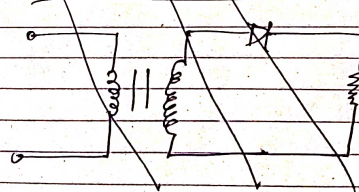
"The process in which AC is converted to DC supply is called rectification process."

"The ckt in which Rectification process takes place is called 'Rectification ckt'."

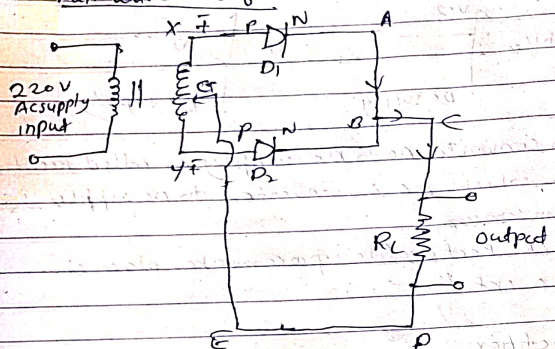
① Half wave Rectifier



② full wave rectifier



full wave Rectifier

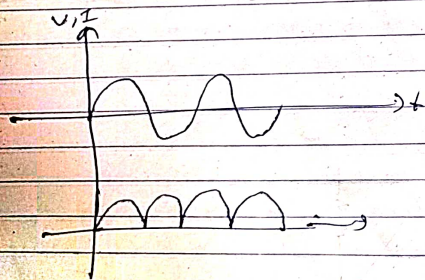


1st half cycle
+ve
 D_1 forward
 D_2 reverse

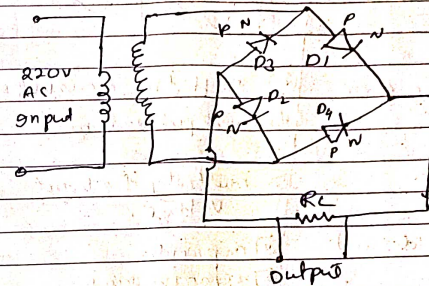
2nd half cycle
-ve
 D_1 reverse
 D_2 forward

1st half cycle
+ve
 D_1 forward
 D_2 reverse

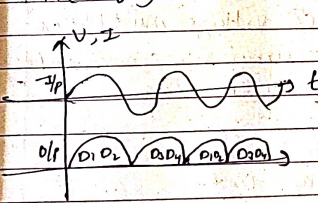
2nd half cycle
-ve
 D_1 reverse
 D_2 forward



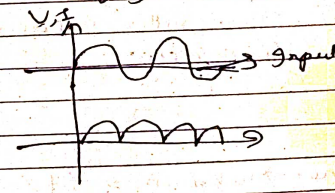
full wave Bridge Rectifier (efficiency and high voltage use then full wave rec.)



+ve half cycle



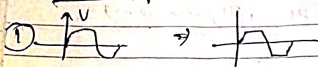
-ve half cycle



zero

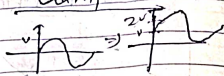
clipper vs clamper

clipper



- 1) Remove unwanted wave form at output voltage
- 2) Less than input voltage
- 3) Nil (energy storage device)
- 4) Slope change
- 5) DC level remains same
- 6) Transmitter, receiver, noise limiter, etc.

clamper



- 1) Shift the DC level of the output voltage
- 2) multiply the input voltage
- 3) capacity
- 4) remains same
- 5) not shifted
- 6) Voltage multiplying circuits, sensor, Radar system etc.

Zener breakdown → It is a phenomenon that occurs in a diode when it is reverse-biased and the electric field across the depletion region becomes strong enough to cause a significant flow of current. ~~It is~~

Under normal reverse biased condⁿ, a diode allows only a small leakage current to pass through it. However, as the reverse voltage across the diode inc, the E within the depletion region inc, when the E reaches a critical value, known as breakdown voltage or Zener voltage, the diode exp. avalanche

and dramatic inc in current flow.

During zener break down, the diode operates in the reverse breakdown region, characterized by a highly non linear current-voltage relationship. Unlike the forward conduction mode of a diode, where the current allows the diode to maintain a nearly constant voltage across its terminal while a large current flows through it.

Zener break down can occur in specially designed diodes known as zener diodes, which are optimized to exhibit to exhibit this effect - zener diodes are heavily doped, allowing them to sustain the high electric fields required for zener break down to occur.

Zener breakdown is a useful phenomenon and finds application in voltage regulation circuits. etc...

Avalanche breakdown → It is a phenomenon that occurs in a diode when it is reverse biased and the electric field across the depletion region causes the generation of charge carriers through collision processes. When a diode is reverse biased, the E within the depletion region inc, if the E is strong enough, it can impart sufficient energy to valence electrons, enabling them to break free from their atomic bonds through collision with other atoms. These liberated e^- → ~~carried~~

Zener breakdown

① occurs in heavily doped diodes designed for controlled breakdown

② E-F causes a sudden inc in current while maintaining nearly constant voltage

③ Lower breakdown voltage.

Avalanche Breakdown

occurs in diodes with lightly doped regions or high doping concn.

E-F causes an exponential inc in current with inc voltage.

Higher breakdown voltage.